EIGHT EQUIVALENT DESCRIPTIONS OF QUASI-COHERENT MODULES ON THE CATEGORY OF AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's own closing aside 'With apologies for the horrible notation' is quoted verbatim as part of the proof.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Quasi-coherence is intrinsic, so the equivalence of ringed sites preserves quasi-coherent modules. Modules on Sites, Lemma sites-modules-lemma-special-locally-free, invoked at proof lines 3779-3782. Transfers quasi-coherence along the equivalence (equation-alternative-ringed).
- Topology-independence of quasi-coherent modules on Sch/S and identification with the O_S-module procedure. Same chapter, Proposition proposition-equivalence-quasi-coherent, invoked at proof lines 3783-3786. Shows (2)-(7) are equivalent and that (8) is the same as (7).
- The sheaf condition for a module determined by localisations on a standard cover. Algebra, Lemma algebra-lemma-cover-module, invoked at proof lines 3814-3815. Gives that F is a Zariski sheaf once the base-change formula is available.
- Compatibility of pullback with the tilde construction, and right exactness of the tensor product. Schemes, Lemma schemes-lemma-widetilde-pullback, invoked at proof lines 3795-3795, and the right exactness of tensor product used at proof lines 3825-3835. Gives (1) implies (2) and the local presentations of F.

Source reference. Original lemma 00EK: sources/stacks-chow/tags.txt, source lines 537–537. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original equation 0GZU: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4727–4727. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 03DX: sources/stacks-descent/pinned-original-source.tex, source lines 2716–2716. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01I9: sources/stacks-chow/tags.txt, source lines 1970–1970. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 03DR: sources/stacks-descent/pinned-original-source.tex, source lines 2355–2355. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03DM: sources/stacks-chow/tags.txt, source lines 4425–4425. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_{Aff}$-modules on $(Aff/S)_{fppf}$. The following are equivalent*

1. *for every morphism $U \to U'$ of $(Aff/S)_{fppf}$ the map $\mathcal{F}(U') \otimes_{\mathcal{O}(U')} \mathcal{O}(U) \to \mathcal{F}(U)$ is an isomorphism,*
2. *$\mathcal{F}$ is a sheaf on $(Aff/S)_{Zar}$ and a quasi-coherent module on the ringed site $((Aff/S)_{Zar}, \mathcal{O}_{Aff})$ in the sense of Modules on Sites, Definition 03DL,*
3. *same as in (2) for the étale topology,*
4. *same as in (2) for the smooth topology,*
5. *same as in (2) for the syntomic topology,*
6. *same as in (2) for the fppf topology,*
7. *$\mathcal{F}$ corresponds to a quasi-coherent module on $(Sch/S)_{Zar}$, $(Sch/S)_{étale}$, $(Sch/S)_{smooth}$, $(Sch/S)_{syntomic}$, or $(Sch/S)_{fppf}$ via the equivalence (0GZU),*
8. *$\mathcal{F}$ comes from a unique quasi-coherent $\mathcal{O}_S$-module $\mathcal{G}$ by the procedure described in Section 03DR.*

Proof. Since the notion of a quasi-coherent module is intrinsic (Modules on Sites, Lemma 03DM) we see that the equivalence (0GZU) induces an equivalence between categories of quasi-coherent modules. Proposition 03DX says the topology we use to study quasi-coherent modules on Sch/S does not matter and it also tells us that (8) is the same as (7). Hence we see that (2) – (8) are all equivalent.

Assume the equivalent conditions (2) – (8) hold and let $\mathcal{G}$ be as in (8). Let $h : U \to U' \to S$ be a morphism of Aff/S. Denote $f : U \to S$ and $f' : U' \to S$ the structure morphisms, so that $f = f' \circ h$. We have $\mathcal{F}(U') = \Gamma(U', (f')^*\mathcal{G})$ and $\mathcal{F}(U) = \Gamma(U, f^*\mathcal{G}) = \Gamma(U, h^*(f')^*\mathcal{G})$. Hence (1) holds by Schemes, Lemma 01I9.

Assume (1) holds. To finish the proof it suffices to prove (2). Let U be an object of $(Aff/S)_{Zar}$. Say $U = \mathrm{Spec}(R)$. A standard open covering $U = U_1 \cup \ldots \cup U_n$ is

given by $U_i = D(f_i)$ for some elements $f_1, \ldots, f_n \in R$ generating the unit ideal of R. By property (1) we see that

$$\mathcal{F}(U_i) = \mathcal{F}(U) \otimes_R R_{f_i} = \mathcal{F}(U)_{f_i}$$

and

$$\mathcal{F}(U_i \cap U_j) = \mathcal{F}(U) \otimes_R R_{f_i f_j} = \mathcal{F}(U)_{f_i f_j}$$

Thus we conclude from Algebra, Lemma 00EK that $\mathcal{F}$ is a sheaf on $(\mathit{Aff}/S)_{Zar}$. Choose a presentation

$$\bigoplus\nolimits_{k \in K} R \longrightarrow \bigoplus\nolimits_{l \in L} R \longrightarrow \mathcal{F}(U) \longrightarrow 0$$

by free R-modules. By property (1) and the right exactness of tensor product we see that for every morphism $U' \to U$ in $(\mathit{Aff}/S)_{Zar}$ we obtain a presentation

$$\bigoplus\nolimits_{k \in K} \mathcal{O}_{Aff}(U') \longrightarrow \bigoplus\nolimits_{l \in L} \mathcal{O}_{Aff}(U') \longrightarrow \mathcal{F}(U') \longrightarrow 0$$

In other words, we see that the restriction of $\mathcal{F}$ to the localized category $(\mathit{Aff}/S)_{Zar}/U$ has a presentation

$$\bigoplus\nolimits_{k \in K} \mathcal{O}_{Aff}|_{(\mathit{Aff}/S)_{Zar}/U} \longrightarrow \bigoplus\nolimits_{l \in L} \mathcal{O}_{Aff}|_{(\mathit{Aff}/S)_{Zar}/U} \longrightarrow \mathcal{F}|_{(\mathit{Aff}/S)_{Zar}/U} \longrightarrow 0$$

With apologies for the horrible notation, this finishes the proof. □